Mev. per fission—can be determined empirically by measuring the velocity of the fragments, or computed from the formula of mass defects (see page 72). Since this is a typically *exothermic* process (a process involving a yielding up of energy), it can take place also without an actual bombardment of the nucleus by neutrons. But the spontaneous fission of nuclei is such a rare phenomenon that technically it has no importance whatever.

Nuclear bombardment by neutrons is capable of producing nuclear fission more or less easily in the different elements at the end of the periodic table (Table III). Certain nuclei can be split even by slow neutrons (i.e. neutrons of thermal velocities). Every one of these is a nucleus with an odd mass number, above all the uranium nucleus $_{92}U^{235}$ and the nucleus of the element plutonium, which will be discussed later. In these nuclei, the small amount of heating linked with the capture of a neutron is sufficient to produce fission. On the other hand, there are other nuclei which can be split by neutrons having a high velocity only. Thus, for instance, neutrons of at least 1 Mev. are required to produce a fission of the uranium nucleus $_{92}U^{238}$. As American experiments with the huge California cyclotron have shown in the course of the past few years, if neutrons (or other nuclear 'projectiles') of more than 30 Mev. are used, it is possible to produce fission in the nuclei at the end of the periodic system as far down as tin.

When nuclei of the latter type are bombarded by slow neutrons, the energy of which is not sufficient to cause fission, the neutrons will be either reflected from the nucleus (mostly with a loss in velocity) or captured (see page 90). Thus, for instance, the nucleus $_{92}U^{238}$ may change by the capture of a neutron into the nucleus $_{92}U^{239}$, which then changes spontaneously, by beta emission, into $_{93}Np^{239}$ (*neptunium*) and $_{94}Pu^{239}$ (*plutonium*). The names 'neptunium' and 'plutonium' were suggested for elements 93 and 94 by the American scientists who first studied the properties of these nuclei. The processes just mentioned can be written as follows:

$$_{92}U^{238} + {}_{0}n^{1} \rightarrow {}_{92}U^{239} \rightarrow {}_{93}Np^{239} + {}_{-1}e^{0}$$
$$_{93}Np^{239} \rightarrow {}_{94}Pu^{239} + {}_{-1}e^{0}$$

The plutonium nucleus, $_{94}Pu^{239}$, thus formed, emits an alpha particle and changes into uranium, $_{92}U^{235}$, although this is a rarer reaction. The half-life of this plutonium nucleus is about 24,000 years. As already explained, the